

TECHNOLOGY

PONTOONS

The military use of floating bridges dates back to ancient times, Persian ruler Xerxes famously building one to cross the Hellespont in 480 BCE. Originally, they were improvised by laying wooden planks over a line of moored boats. By the 19th century, however, army engineers had purpose-built flat-bottomed pontoons as part of their standard equipment. Mounted on wheels for ease of mobility, they were threaded together on a cable with wooden beams laid across them to form a roadway.



Burnside's advance

The bulk of the Union Army began marching over the pontoon bridges on December 12, many soldiers looting the abandoned Virginian homes. The battle was fought on December 13. Burnside's best hope rested on Major General William Franklin's Grand Division, which had crossed the river south of Fredericksburg. Burnside thought that the Confederate right flank in front of Franklin was weakly held, but Jackson had amassed his forces around the peak of Prospect Hill. When a Confederate officer expressed anxiety about their situation, Jackson put him firmly in his place. He was proved right, because the Union troops' frontal assaults on his well-placed infantry and cannon were systematically repulsed. Startled early on by an unexpected flank attack from Confederate horse artillery, Franklin cautiously held many of his soldiers back in defensive positions.

The breakthrough occurred when a rush through an undefended wooded valley by Major General George Meade's troops penetrated deep into the Confederate army's lines. But Franklin failed to reinforce Meade, and his men were soon driven back by counterattacks and suffered heavy losses as a result. Nothing on Jackson's flank, however, equaled the appalling slaughter inflicted by Longstreet's corps at Marye's Heights. Union brigades were thrown forward in frontal assaults uphill across open ground swept